

Functional Annotation Report: PP_4594 (Q88E72), *Pseudomonas putida* KT2440

Gene: PP_4594 (OrderedLocusNames PP_4594) **UniProt:** Q88E72 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440) — proteome code PSEPK **Submitted name:** Cystathionine gamma-synthase **Protein family:** Trans-sulfuration enzymes family (Cys/Met metabolism, PLP-dependent, fold-type I "γ-family")

Summary

PP_4594 encodes a **393-residue, cytoplasmic, pyridoxal-5'-phosphate (PLP)-dependent enzyme** belonging to the **fold-type I Cys/Met-metabolism "γ-family"** of trans-sulfuration enzymes (Pfam **PF01053**, Cys_Met_Meta_PP; InterPro **IPR000277**). Its sequence carries a **fully intact catalytic apparatus** — the Schiff-base lysine (Lys205) that forms the internal aldimine with PLP, and the conserved aspartate (Asp180) that anchors the PLP pyridinium nitrogen — so the protein is very likely a **catalytically active PLP enzyme**, not a pseudoenzyme. Enzymes of this family carry out **α,γ-elimination and γ-replacement reactions on sulfur-containing amino acids** (cystathionine, homocysteine, methionine and their O-substituted derivatives) and share a common three-domain, homotetrameric fold.

Assigning the **precise reaction** catalyzed by PP_4594 is the central challenge of this annotation, because two credible calls exist and no direct enzymology has ever been performed on the PP_4594 protein itself. The **original 2002 genome annotation and PIRSF/CDD signatures name it "cystathionine γ-synthase"**, whereas **KEGG assigns it the orthology K01758 = cystathionine γ-lyase (EC 4.4.1.1)**. The weight of convergent bioinformatic evidence favors the **γ-lyase-type function**: PP_4594 falls into the **same PANTHER subfamily (PTHR11808:SF15)** as the experimentally characterized *P. aeruginosa* cystathionine γ-lyase PA0400, its **closest characterized homolog at ~47% identity**, and in a pairwise-identity/phylogenetic comparison it clusters with the **γ-elimination clade** (human CTH 42%, *P. putida* methionine γ-lyase 42%) and is clearly separated from the β-lyase clade (E. coli MetC 28%, KT2440 MetC/PP_4348 24%). Importantly, PP_4594 is **distinct from KT2440's dedicated forward methionine-biosynthetic genes metB (PP_0659) and metC (PP_4348)**, and

it sits immediately adjacent to a **cysteine-sensing Lrp/AsnC-family transcriptional activator (PP_4595)** — a genomic arrangement consistent with a cysteine-inducible, accessory catabolic role rather than a housekeeping biosynthetic one.

The most defensible functional statement is therefore: **PP_4594 is a cytoplasmic PLP-dependent γ -family trans-sulfuration enzyme, most probably acting as a cystathionine γ -lyase (γ -elimination on cystathionine/homocysteine-type substrates, potentially producing H_2S), likely functioning as an accessory, cysteine-inducible enzyme in sulfur-amino-acid metabolism.** This assignment rests on orthology and sequence/structure inference; the exact substrate specificity and reaction direction remain to be confirmed by direct biochemical characterization. A significant caveat is that Vermeij & Kertesz (1999) could **not detect cystathionine γ -lyase activity in *P. putida***, which tempers over-confident acceptance of the K01758 label.

Key Findings

Finding 1 — PP_4594 is a cytoplasmic PLP-dependent trans-sulfuration (Cys/Met γ -family) enzyme

UniProt entry Q88E72 describes a **393-amino-acid protein** whose cofactor is **pyridoxal 5'-phosphate (PLP)** and whose family membership is annotated as "Belongs to the trans-sulfuration enzymes family." This placement is supported by a dense, mutually consistent set of domain signatures:

Signature	Database	Meaning
PF01053 (Cys_Met_Meta_PP)	Pfam	Cys/Met metabolism PLP-dependent enzyme
IPR000277 (Cys/Met-Metab_PyrdxLP-dep_enz)	InterPro	Cys/Met metabolism, PLP-dependent
IPR015421 / IPR015422 / IPR015424	InterPro	PLP-dependent transferase, fold-type I (major/small/overall)
SSF53383	SUPFAM	PLP-dependent transferases superfamily
COG0626	eggNOG	Cystathionine β -lyase / γ -synthase family

Gene Ontology terms assigned to the entry are **pyridoxal phosphate binding (GO:0030170)**, **cytoplasm (GO:0005737)**, and **transsulfuration (GO:0019346)**. All of these localization/function terms are currently **IEA (inferred from electronic annotation)**, meaning they are computational rather than experimentally verified for this specific protein. Nonetheless, the convergence of independent domain databases makes the **fold assignment (fold-type I PLP transferase) and the broad functional category (Cys/Met γ -family trans-sulfuration enzyme) highly secure**. The predicted cytoplasmic localization is fully consistent with the biology of this enzyme family, whose members are soluble cytosolic proteins with no signal peptides or membrane-spanning segments.

Finding 2 — PP_4594 is annotated as cystathionine γ -lyase (K01758 / EC 4.4.1.1) and is distinct from KT2440's dedicated forward-transsulfuration genes

There is a genuine **annotation conflict** at the heart of this gene. KEGG assigns `ppu:PP_4594` to ortholog **K01758 (cystathionine γ -lyase, EC 4.4.1.1)**, whereas the original 2002 genome submission and the UniProt "SubName" call it **cystathionine γ -synthase**. Resolving this required situating PP_4594 relative to the other Cys/Met enzymes encoded in the KT2440 genome and relative to characterized reference enzymes.

In KT2440 the canonical forward methionine-biosynthetic genes are **separate loci**: - **metB (cystathionine γ -synthase, K01739) = PP_0659** - **metC (cystathionine β -lyase, K01760) = PP_4348** - **methionine γ -lyase (K01761) = PP_1308**

Because these dedicated biosynthetic functions are already accounted for elsewhere in the genome, PP_4594 is **not** the housekeeping γ -synthase or β -lyase of methionine biosynthesis; it is an additional, paralogous γ -family enzyme.

An independent Needleman–Wunsch pairwise-identity analysis places PP_4594 firmly in the **γ -elimination clade**:

Reference enzyme	% identity to PP_4594	Clade
Human CGL/CTH (cystathionine γ -lyase)	42.4%	γ -lyase
<i>P. putida</i> methionine γ -lyase (PpMGL)	41.7%	γ -family
Yeast CGL/CYS3	40.5%	γ -lyase
<i>E. coli</i> MetB (γ -synthase)	37.6%	γ -synthase
KT2440 MetB / PP_0659	37.4%	γ -synthase
<i>E. coli</i> MetC (β -lyase)	28.4%	β -lyase
KT2440 MetC / PP_4348 (β -lyase)	23.7%	β -lyase

PP_4594 is **most similar to γ -lyases and clearly separated from the β -lyase clade**, supporting a γ -elimination (γ -lyase / γ -family) rather than a β -elimination function. Its genomic neighbor, **PP_4595, is a Lrp/AsnC-family cysteine-sensing transcriptional activator (K05800)** — a regulatory context that strongly suggests PP_4594 expression is **cysteine-responsive**.

Finding 3 — Physiological context: *P. putida* uses direct sulfhydrylation for methionine synthesis; transsulfuration enzymes are low-level and cysteine-inducible

The physiology of sulfur metabolism in *P. putida* was directly measured by Vermeij & Kertesz (1999) [PMID: 10482527](#). Their findings frame the likely role of PP_4594:

- *P. putida* (and *P. aeruginosa*) **"used direct sulfhydrylation of O-succinylhomoserine for the synthesis of methionine"** and also contained substantial O-acetylserine sulfhydrylase (cysteine synthase) activity. This means the **main methionine-biosynthetic route bypasses the transsulfuration enzymes**, casting them as accessory rather than essential.
- **"The enzymes of the transsulfuration pathway (cystathionine gamma-synthase and cystathionine beta-lyase) were expressed at low levels in both pseudomonads but were strongly upregulated during growth with cysteine as the sole sulfur source."** This matches the presence of the cysteine-sensing Lrp/AsnC regulator (PP_4595) beside PP_4594 and supports a **cysteine-inducible** expression pattern.
- Critically, **"P. putida S-313 also grew well with methionine as the sulfur source, but no cystathionine gamma-lyase, the key enzyme of the reverse transsulfuration pathway,**

was found in this species" — instead, methionine was degraded via methionine γ -lyase.

This is important **counter-evidence** that cautions against uncritically accepting the KEGG K01758 (cystathionine γ -lyase) call for PP_4594 as its principal *in vivo* activity.

Structural context reinforces the shared fold. Motoshima et al. (2000) [PMID: 10965031](#) showed that *P. putida* methionine γ -lyase adopts a fold in which "the spatial fold of subunits, with three functionally distinct domains and their quarternary arrangement, is **similar to those of L-cystathionine beta-lyase and L-cystathionine gamma-synthase from Escherichia coli**," confirming the common fold-type I, three-domain architecture across the entire γ -family to which PP_4594 belongs.

Finding 4 — PP_4594 has a fully intact fold-type I PLP catalytic apparatus (Schiff-base Lys205, PLP-anchoring Asp180)

Motif mapping of the 393-residue Q88E72 sequence identified **all signature catalytic elements** of the fold-type I Cys/Met γ -family, indicating the protein is a **catalytically competent enzyme**, not a degenerate pseudoenzyme:

Catalytic element	Residues / motif	Role
Schiff-base lysine	Lys205 in H-S-A-S-K-Y (201–206)	Forms internal aldimine with PLP; central to catalysis
Conserved aspartate	Asp180 in D-N-T-F (180–183)	Hydrogen-bonds/orients the PLP pyridinium nitrogen
PLP phosphate-binding loop	E-S-P-T-N-P (151–156)	Binds the PLP phosphate group
N-terminal substrate/cofactor loop	G-N-P-T-R (57–61)	Substrate and cofactor positioning

Additional structural classifiers agree: **CDD cd00614 (CGS_like)**, **PIRSF001434 (Cystathionine gamma-synthase)**, and **CATH 3.40.640.10 / 3.90.1150.10** (aspartate aminotransferase-like two-domain PLP fold). This is the canonical architecture shared by cystathionine γ -synthase, cystathionine γ -lyase, and methionine γ -lyase (Motoshima 2000). The presence of a complete, unmutated catalytic set means the enzyme **retains the machinery to perform PLP-dependent α,γ -eliminations/ γ -replacements** on appropriate sulfur-amino-acid substrates.

Finding 5 — PP_4594 co-classifies with, and is the closest characterized homolog of, the *P. aeruginosa* cystathionine γ -lyase PA0400 — favoring a γ -lyase over a γ -synthase call

The single most informative comparison for narrowing PP_4594's function is to **PA0400 (Q9I6A0) of *P. aeruginosa* PA01**, annotated as "probable cystathionine γ -lyase" and the subject of a 2024 structure/specificity study, Pedretti et al. [PMID: 38654065](#). PP_4594 and PA0400:

- are assigned to the **same PANTHER subfamily, PTHR11808:SF15**, and
- share **47.4% pairwise identity — higher than to any other characterized reference** (human CTH 42.4%, PpMGL 41.7%, *E. coli* MetB 37.6%, *E. coli* MetC β -lyase 28.4%).

Because PA0400 is a *characterized* enzyme placed in **bacterial defense against oxidative stress and antibiotics via H₂S production**, and because PP_4594 is its nearest characterized relative in the same fine-grained subfamily, the balance of evidence favors a **cystathionine γ -lyase-type (γ -elimination, potentially H₂S-generating) function** for PP_4594. The Pedretti et al. study frames this defense role explicitly: "The escalating drug resistance among microorganisms underscores the urgent need for innovative therapeutic strategies and a comprehensive understanding of bacteria's defense mechanisms against oxidative stress and antibiotics." KEGG's independent assignment of K01758 to PP_4594 is consistent with the γ -lyase call.

The **counter-evidence for a synthase call** should be stated plainly: the original 2002 genome name, the PIRSF001434 label ("Cystathionine gamma-synthase"), and the CDD cd00614 name ("CGS_like") all describe the broader clade as γ -synthase-like, and PP_4594's identity to true γ -synthases (*E. coli* MetB 37.6%; KT2440 metB/PP_0659 37.4%) is only modestly lower than its identity to γ -lyases. The γ -synthase vs. γ -lyase distinction within this fold can hinge on a small number of active-site residues, so **the assignment is a probabilistic lean, not a certainty**.

Mechanistic Model / Interpretation

The enzyme family and its chemistry

PP_4594 belongs to the **fold-type I (aspartate-aminotransferase-like) PLP enzyme superfamily**, specifically the **Cys/Met-metabolism "γ-family"** (trans-sulfuration enzymes). Members are typically **homotetramers** in which each active site is built from residues of two adjacent subunits, with a bound PLP cofactor tethered as an internal aldimine (Schiff base) to the catalytic lysine (here **Lys205**). Substrate amino acids displace this lysine to form an external aldimine, after which PLP stabilizes carbanionic intermediates enabling elimination or replacement chemistry at the Cβ or Cγ position.

Cys/Met γ-family reactions on sulfur amino acids
(fold-type I PLP enzymes; PP_4594 belongs here)

γ-synthase (metB): 0-succinyl-L-homoserine + L-cysteine
→ L-cystathionine + succinate (γ-replacement)

β-lyase (metC): L-cystathionine + H₂O
→ L-homocysteine + pyruvate + NH₃ (β-elimination)

γ-lyase (CTH): L-cystathionine + H₂O
→ L-cysteine + 2-oxobutanoate + NH₃ (γ-elimination)
(side reactions can release H₂S from Cys/homocysteine)

Met γ-lyase(MGL): L-methionine + H₂O
→ methanethiol + 2-oxobutanoate + NH₃ (α,γ-elimination)

All four activities are performed by structurally near-identical enzymes; **substrate specificity and reaction type are dictated by a small set of active-site residues**, which is precisely why sequence-only annotation is ambiguous here. Structural studies of *P. putida* MGL (e.g., [PMID: 28329912](#), [PMID: 34953671](#)) demonstrate that single active-site substitutions can switch substrate preference between methionine and homocysteine, underscoring the limits of homology-only calls.

Where PP_4594 sits

KT2440 sulfur-amino-acid enzyme inventory

PP_0659	metB	cystathionine γ -synthase	K01739	(forward Met biosynth)
PP_4348	metC	cystathionine β -lyase	K01760	(forward Met biosynth)
PP_1308	---	methionine γ -lyase	K01761	(Met catabolism)
PP_4594	???	" γ -synthase"/ γ -lyase	K01758	← THIS GENE
PP_4595	---	Lrp/AsnC Cys-sensing activator	K05800	(neighbor/regulator)

Pairwise identity places PP_4594 in the γ -LYASE clade, closest to *P. aeruginosa* PA0400 (47%) – a characterized γ -lyase.

The presence of dedicated metB and metC elsewhere means PP_4594 is a **surplus γ -family paralog**. Combined with (a) its clustering with γ -lyases, (b) its ~47% identity to the characterized *P. aeruginosa* γ -lyase PA0400 in the same PANTHER subfamily, and (c) its cysteine-sensing regulatory neighbor PP_4595, the coherent model is:

PP_4594 is a cytoplasmic, cysteine-inducible, accessory PLP-dependent γ -family enzyme most likely performing a cystathionine γ -lyase-type γ -elimination, contributing to sulfur-amino-acid interconversion/catabolism (with possible H₂S generation), rather than serving as the primary methionine-biosynthetic γ -synthase.

The one loose thread is Vermeij & Kertesz's failure to detect classical *reverse-transsulfuration* cystathionine γ -lyase activity in *P. putida*. This can be reconciled if PP_4594's true in vivo activity is a related γ -elimination (e.g., on homocysteine or a non-canonical substrate) that was not assayed, or if the enzyme is expressed only under conditions not tested in that study — but it does mean the **K01758 = "cystathionine γ -lyase" label should be treated as a best-available hypothesis, not an established fact.**

Evidence Base

PMID	Title (abbrev.)	How it supports / challenges the annotation
10482527	<i>Pathways of assimilative sulfur metabolism in P. putida</i>	Supports & tempers. Establishes direct sulfhydrylation as the main Met route (transsulfuration is accessory); shows transsulfuration enzymes are cysteine-inducible (matches PP_4595 neighbor). But reports cystathionine γ -lyase activity was <i>not detected</i> in <i>P. putida</i> — key caveat against the K01758 call.
10965031	<i>Crystal structure of PLP-dependent L-methionine γ-lyase from P. putida</i>	Supports fold. Confirms shared fold-type I, three-domain architecture across γ -family enzymes (β -lyase, γ -synthase, γ -lyase, MGL), validating PP_4594's structural class.
38654065	<i>Catalytic specificity and crystal structure of cystathionine γ-lyase from P. aeruginosa</i>	Supports γ-lyase call. PA0400 is PP_4594's closest characterized homolog (47%, same PANTHER SF15); places bacterial γ -lyase in oxidative-stress/antibiotic defense via H ₂ S, suggesting an analogous role for PP_4594.
8586629	<i>Structural analysis of the L-methionine γ-lyase gene from P. putida</i>	Documents the extensive homology of MGL with cystathionine γ -lyase and γ -synthase — the same α,γ -elimination/ γ -replacement PLP enzyme group PP_4594 belongs to.
26913973	<i>Revisited genome of P. putida KT2440</i>	Provides the re-annotation framework for KT2440 gene functions used to distinguish PP_4594 from metB/metC.
28329912 , 34953671	<i>Mechanistic / mutant studies of P. putida MGL</i>	Illustrate how single active-site substitutions switch substrate preference (Met $\leftrightarrow$ homocysteine) in this fold — explaining why γ -synthase vs. γ -lyase cannot be settled by sequence alone.
39427081	<i>Reverse transsulfuration pathway and colitis</i>	Defines cystathionine γ -lyase (CTH) chemistry (cystathionine $\rightarrow$ cysteine) — the reaction KEGG proposes for PP_4594.

Confidence in fold/family assignment: very high. Multiple independent domain databases (Pfam, InterPro, SUPFAM, CDD, CATH, eggNOG) concur, and all catalytic residues are intact.

Confidence in precise reaction (γ -lyase vs. γ -synthase): moderate. The lean is toward γ -lyase (KEGG K01758; closest to characterized PA0400; γ -clade clustering), but the original genome/PIRSF/CDD names say γ -synthase, and no direct assay exists.

Limitations and Knowledge Gaps

- 1. No direct enzymology on the PP_4594 protein.** Every functional statement here is inferred from orthology, domain signatures, and sequence/structure comparison. The substrate specificity (cystathionine? homocysteine? O-substituted homoserine?), reaction direction (elimination vs. replacement), and kinetic parameters have never been measured for Q88E72.
 - 2. Annotation conflict is unresolved.** KEGG (γ -lyase, K01758) vs. original genome/PIRSF/CDD (γ -synthase). Within this fold, specificity can hinge on a few active-site residues, so bioinformatics cannot fully settle it.
 - 3. All GO localization/function terms are IEA (computational).** No experimental localization data confirm the predicted cytoplasmic site (though it is biologically near-certain for a soluble PLP enzyme with no targeting signals).
 - 4. Contradictory physiology.** Vermeij & Kertesz did not detect cystathionine γ -lyase activity in *P. putida*, in tension with the K01758 assignment — possibly a condition- or substrate-dependent gap rather than a true contradiction.
 - 5. Regulatory linkage is inferred, not proven.** PP_4595 is a cysteine-sensing Lrp/AsnC activator by annotation; direct evidence that it controls PP_4594 (operon structure, ChIP, promoter mapping) has not been established.
 - 6. H₂S-generating / defense role is by analogy only** to *P. aeruginosa* PA0400 and has not been tested in *P. putida*.
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Proposed Follow-up Experiments / Actions

- 1. Heterologous expression and enzymatic assay.** Clone PP_4594, purify the His-tagged protein, confirm PLP loading (absorbance at ~420 nm), and assay candidate reactions: cystathionine γ -elimination ($\rightarrow$ cysteine + 2-oxobutanoate + NH₃), cystathionine γ -synthesis (O-succinyl/acetyl-homoserine + cysteine), methionine/homocysteine γ -elimination, and H₂S

release (lead-acetate or fluorescent probe). Report k_{cat}/K_m for each substrate to settle the γ -lyase-vs- γ -synthase question definitively.

2. **Crystal structure or AlphaFold-model active-site comparison.** Solve/model the structure and superpose the active site onto PA0400 (γ -lyase) vs. *E. coli* MetB (γ -synthase) to identify the specificity-determining residues and predict substrate scope.
3. **Transcriptional regulation test.** Fuse the PP_4594 promoter to a reporter and measure induction under cysteine vs. sulfate vs. methionine as sole sulfur source; delete PP_4595 to test whether it is the activating regulator, confirming or refuting the cysteine-inducible model.
4. **Phenotypic characterization of a knockout.** Construct a Δ PP_4594 mutant and test growth on various sulfur sources, sensitivity to oxidative stress/antibiotics (the PA0400 defense hypothesis), and endogenous H_2S levels.
5. **Metabolomic tracing.** Use ^{34}S - or ^{13}C -labeled cysteine/cystathionine to trace flux through PP_4594 in vivo, distinguishing forward biosynthetic from catabolic/reverse-transsulfuration roles.

Conclusion

PP_4594 (Q88E72) of *Pseudomonas putida* KT2440 is a **cytoplasmic, 393-residue, PLP-dependent fold-type I enzyme of the Cys/Met-metabolism γ -family (trans-sulfuration family)** with an intact catalytic apparatus (Schiff-base Lys205, PLP-anchoring Asp180). Convergent bioinformatic evidence — KEGG ortholog K01758, shared PANTHER subfamily PTHR11808:SF15 with the characterized *P. aeruginosa* cystathionine γ -lyase PA0400 (its closest characterized homolog at ~47% identity), and clustering with the γ -elimination clade — indicates its **most probable activity is a cystathionine γ -lyase-type γ -elimination (EC 4.4.1.1)** acting on cystathionine/homocysteine-type sulfur amino acids in the cytosol, possibly generating H_2S . It is **distinct from KT2440's dedicated forward methionine-biosynthetic genes metB (PP_0659) and metC (PP_4348)** and is most consistent with an **accessory, cysteine-inducible catabolic role** (adjacent to the cysteine-sensing Lrp regulator PP_4595). This assignment is inference-based: the original genome/PIRSF annotation calls it "cystathionine γ -synthase," no direct enzymology has been performed on the protein, and *P. putida* was reported to lack detectable cystathionine γ -lyase activity — so the exact substrate and reaction direction await experimental confirmation.

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